

Roohul Pandith



Contact

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Education

- **MBA - 2024**
Subharti University
70.00%
- **B.Tech (ECE) - 2016**
Sharda University
8.7 CGPA
- **XIIth -2011**
GBHS (JKBOSE)
80.40 %
- **Xth -2009**
GBHS (JKBOSE)
89.80 %

Professional Summary

Electric Powertrain and HV Battery Systems Engineer with 9+ years of international and cross-functional experience across top OEMs including Jaguar Land Rover (UK), VinFast (Vietnam), and Maruti Suzuki (India). Adept at full lifecycle management of EV systems—from architecture, control algorithm design, and state flow modeling to diagnostics, validation, and issue resolution. Skilled at translating high-level requirements into robust technical deliverables, ensuring seamless system integration, supplier alignment, and production readiness. Known for a structured approach to problem-solving and driving innovation in embedded system behavior, diagnostics, and e-powertrain control logic.

Key work abilities and Skills

- High Voltage Battery System Design, Validation & System Benchmarking
- Battery Management System (BMS) Architecture, Functional Algorithms & Safety Logic
- Electric Powertrain Systems Engineering: Architecture, Requirements, Swim lanes & State Machines
- HV Battery & Charging System : DFMEA, DVP&R, Technical Requirement Specifications (SOR)
- CAN Protocols (ISO 11898), UDS Diagnostics (ISO 14229), CANalyzer, Canoe
- UDS Flashing, Calibration Testing & Functional behavior Verification
- Signal Logging, DTC behavior Testing, and Fault Tracing using Vector & ETAS Tools
- Integration Testing in MIL, HIL, and Full Vehicle Environments
- Root Cause Analysis & Issue Management: 8D, FTA, Why-Why Analysis
- Software Defect Management, Change Tracking & Supplier Coordination
- Client Program Management & Delivery

Professional Experience

Technical Lead - ePowertrain - 02/2023 till Present

Tata Technologies, Pune-India

Jaguar Land Rover, UK

- Delivered comprehensive HV System Engineering outputs including DVP, DFMEA, system swim lanes, and architecture specifications.
- Authored state flow and fault logic documentation for charger, inverter, and power distribution systems.
- Led technical analysis and resolution of powertrain and charging-related issues across EMA/MLA platforms.
- Collaborated with cross-functional teams to support proto build activities, production validation, and service flow creation.
- Contributed to software integration testing, reviewing system interactions and ensuring logic consistency across modules.

Agratas, India

- Spearheaded HV Battery System Engineering deliverables including detailed DVP, DFMEA, and BMS requirement validations.
- Conducted competitive benchmarking of battery architecture and performance for next-gen platforms.

Skills

- Matlab/Simulink
- JIRA, Share Point & Lucid.
- Canoe, Canalyzer, vSignalyzer.
- INCA and MDA.
- DataSpy
- Wivi, Beatrice, TB2
- Hioki & Keyence loggers.
- Vector GL series Loggers.
- Chroma Battery Cycler.
- Renesas Flashing tool
- Perch UDS.
- dSpace Micro-Autobox.
- dSpace Control Desk.
- RCA, 8D etc.
- Technical roadmap planning

Certifications

- C.Eng(I) : M1816696
- Master Project management (Udemy Course)
- Embedded System
- Verilog-HDL

Vinfast, Vietnam

- Directed technical ownership for ePowertrain components: HV Battery, CPDU, VCU, and e-Shifter on VF6/VF7 programs.
- Developed detailed state machines, regenerative braking logic, drive control algorithms, and charger flow sequences.
- Interfaced with software and system suppliers to track software releases, validate bug fixes, and close issue tickets.
- Reviewed system architecture against vehicle performance targets and supported final EE sign-off through system integration testing.

Deputy Manager: HV Battery- 06/2018 to 02/2023

Maruti Suzuki India Limited, Gurgaon-India

- Designed and implemented advanced BMS control strategies for SOC/SOH estimation, thermal control, passive balancing, and fault management.
- Developed high-fidelity plant models in MATLAB/Simulink and executed MIL and HIL testing using dSPACE MicroAutoBox and Control Desk.
- Authored and validated UDS diagnostic routines in accordance with ISO 14229 standards; ensured complete DTC matrix coverage.
- Executed software flashing and UDS calibration verification using Suzuki tools; supported software bring-up and issue debugging.
- Investigated hybrid 12V and HV battery issues from the field; conducted structured root cause analyses and supported countermeasure deployment.
- Led component integration, test planning, bench and vehicle-level validation for battery systems.
- Worked with suppliers and internal validation teams to establish DVPs and manage software test execution.
- Simulated and tested charging/discharging strategies, contributing to drive mode calibration and energy optimization.
- Reviewed and maintained TR documents, DFMEA sheets, and test traceability matrices to ensure audit readiness and compliance.

Software Engineer - 11/2016 to 05/2018

Aon Hewitt, Gurgaon-India

- Managed end-to-end development, customization, and deployment of Salesforce CRM modules.
- Designed and maintained user roles, workflows, validation rules, dashboards, and data migration processes.
- Conducted unit testing and configuration across sandbox environments; supported client delivery and production deployments.

Patents & Innovations

- Granted Patent #565222 – Method for detecting unauthorized opening of traction battery pack
- Granted Patent #501237 – System for detecting abnormality of passive protection devices in electric vehicles
- Pioneered diagnostic innovation and algorithm optimization across multiple projects, leading to granted IP.

Declaration:

I, Roohul A. Pandith hereby declare that the information compiled above is authentic and genuine.